

End Semester Examination

May-June 2023

CET0004B - Algorithms and Data Structures

Schedule ID: 19327

Faculty/School	Faculty of Engineering and Technology	Term	Semester III
Program	SY CSE	Duration	1 Hours 30 Minutes
Specialization	--	Max. Marks	40

Read the instructions provided for every question properly before attempting the answer.

Section - 1 : contain(s) **10** question(s) and each question carries **5** mark(s). You can answer any **8** questions out of **10**.

Click **Finish** only after completion of the Exam.

Section - 1 (8 X 5 Marks) Answer any 8 questions

1	Define data structure. Explain Classification of data structure in detail.	5 marks	CO1	Remembering
2	Define the term algorithm. List & Explain different approaches to design an algorithm.	5 marks	CO1	Remembering
3	Explain time and space complexity with an example.	5 marks	CO2	Understanding
4	Write a 'C' program to find factorial of positive integer N.	5 marks	CO2	Applying
5	Find the position of element 21 using Binary Search method in Array 'A' given below : A = {11, 5, 21, 3, 29, 17, 2, 45}	5 marks	CO3	Applying
6	Describe working of bubble sort with example.	5 marks	CO3	Understanding
7	Evaluate following postfix expressions show details of stack step by step 5,6,2,+,*,12,4,/,-	5 marks	CO4	Evaluating
8	Write algorithm for performing push and pop operations on stack.	5 marks	CO4	Remembering
9	Explain the concept of circular queue in detail with examples.	5 marks	CO5	Understanding
10	Write a program in 'C' to insert an element in a linear queue.	5 marks	CO5	Applying

END OF QUESTION PAPER